PERSONAL RENTAL FINDER · DUBAI & ABU DHABI

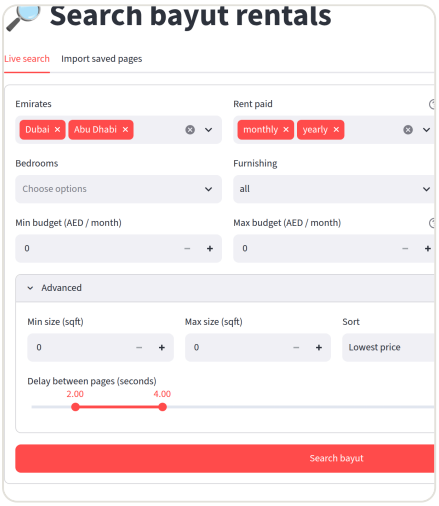
Bayut Finder

A step-by-step visual guide to the rebuilt app: how each filter becomes a Bayut search, how listings become numbers, what to watch out for, and where to take it next.

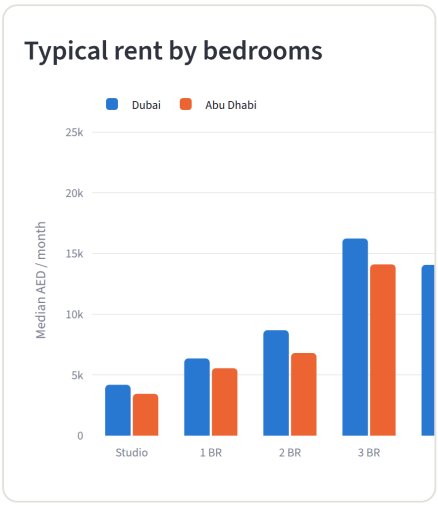
Version 2 · October 2026

Streamlit app + Python package

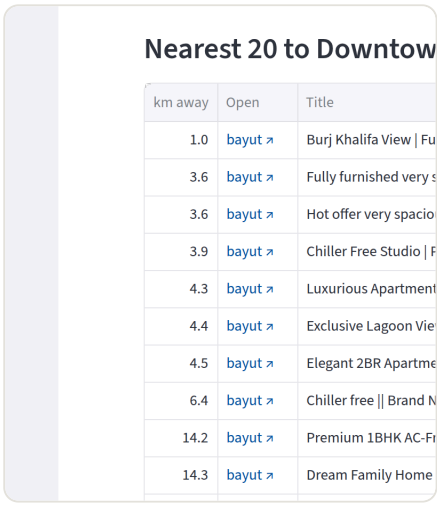
Branch `claude/affectionate-hopper-0imoq0`



Search: pick filters



Compare: rents side by side



Map: nearest listings

1 · Search

Emirates, monthly or yearly, type, beds, budget. The app pages through bayut.com for you.

2 · Normalise

Every listing gets AED / month *and* AED / year, so monthly and yearly prices compare fairly.

3 · Compare

Cheapest communities, rent by bedrooms, the true cost of paying monthly, below-median deals.

4 · Locate

See listings on a map and rank them by distance to your office or any place.

Screenshots marked **REAL DATA** come from the bayut results page you saved (24 Dubai listings, 3 Oct 2026). Screenshots marked **SAMPLE DATA** use a generated demo dataset, so every chart has something to show; run your own search for real numbers.

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CHAPTER 1

The app at a glance

Bayut Finder answers one question: **where can I rent, for how much, and is this listing a fair price?** It collects Bayut rental listings for Dubai and Abu Dhabi, puts every price on the same monthly and yearly scale, and lays them out so you can compare areas and spot good deals.

24
listings per Bayut results page. Pages are fetched one by one, politely spaced.

20+ fields
per listing: price, period, beds, baths, size, GPS, community, building, agency, link…

0 API keys
GPS comes from the page itself; maps and place search use free OpenStreetMap.

The 4-step routine

1 **Search**
Open **Search**, choose emirates, rent period (monthly and/or yearly), property type, bedrooms and a **monthly** budget, then click **Search bayut**. Results are saved automatically to `data/`.

2 **Compare**
Open **Compare** to see typical rent by bedrooms, the cheapest communities, how much more monthly listings cost over a year, and a shortlist of listings priced below similar ones.

3 **Locate**
Open **Map**, pick a landmark (e.g. DIFC, Yas Island) or type any place, set a radius, and get the nearest listings with distances.

4 **Decide**
Click **bayut ↗** in any table to open the original listing, contact the agent, and re-run the search a few days later to see what changed.

What changed from the old version

Topic	Old notebooks (2023–2025)	Rebuilt app (2026)
Finding data on the page	Hashed CSS classes like `_475e888a`, which break on every Bayut update	Stable `aria-label`s + the page's structured JSON-LD data
Getting the page	`requests`, now blocked (HTTP 403)	`curl_cffi` impersonating Chrome; browser and import fallbacks
Coordinates	Google Maps geocoding with an API key, one call per listing	Read directly from the page, per listing, no API key
Monthly vs yearly	Separate, not comparable	Both columns for every listing + premium analysis
URL building	String concatenation; a second `?` broke combined filters	One builder that encodes every filter correctly
Failures & quality	Silent empty DataFrame; no tests	Clear message + raw page in `data/debug/`; 18 offline tests

CHAPTER 2

Run it from Anaconda Prompt

Your packages are already installed, so running the app is three commands. The app opens in your browser at **http://localhost:8501**.

1 Go to the project folder (the one that contains `Home.py`)

```
ANACONDA PROMPT
(base) C:\> cd /d C:\path\to\bayutfinder
:: /d also switches drive, e.g. to D:\
```

2 Activate your environment (skip if you use `base`)

```
(base) C:\path\to\bayutfinder> conda activate base
```

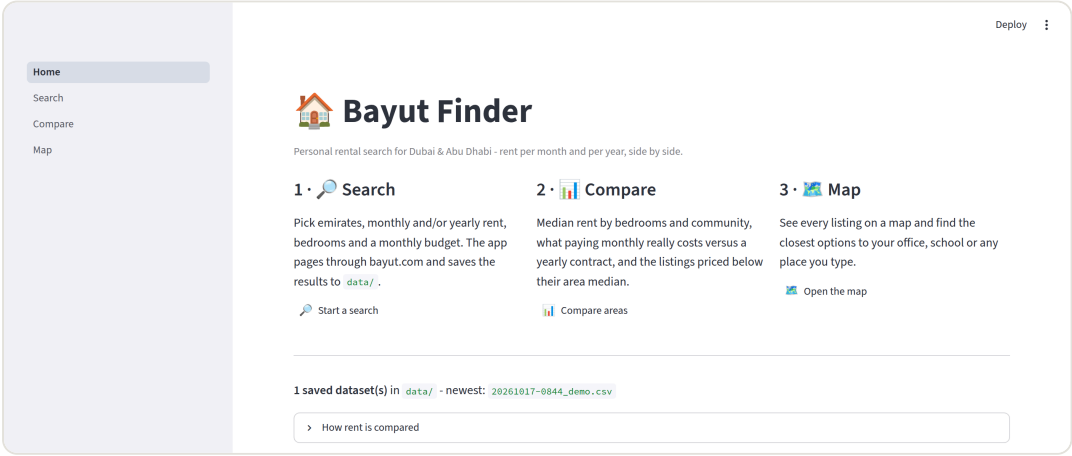
3 Start the app

```
(base) C:\path\to\bayutfinder> streamlit run Home.py
  You can now view your Streamlit app in your browser.
  Local URL: http://localhost:8501
:: stop the app with Ctrl+C in this window
```

⚠ Version check, if a page shows an error

The app needs **streamlit ≥ 1.50** and **plotly ≥ 5.24**, plus **curl_cffi**. Older Anaconda installs give errors like `unexpected keyword argument 'width'` or `no attribute 'pills'`.

```
> python -c "import streamlit, plotly, curl_cffi; print(streamlit.__version__,
plotly.__version__)"
> pip install -U "streamlit≥1.50" "plotly≥5.24" curl_cffi
```



APP The Home page: three cards link to Search, Compare and Map; the sidebar lists the same pages.

CHAPTER 3

How it works end to end

Every search follows the same pipeline. Each stage is one small Python module in `bayut_finder/`, so when something breaks you know exactly where to look.

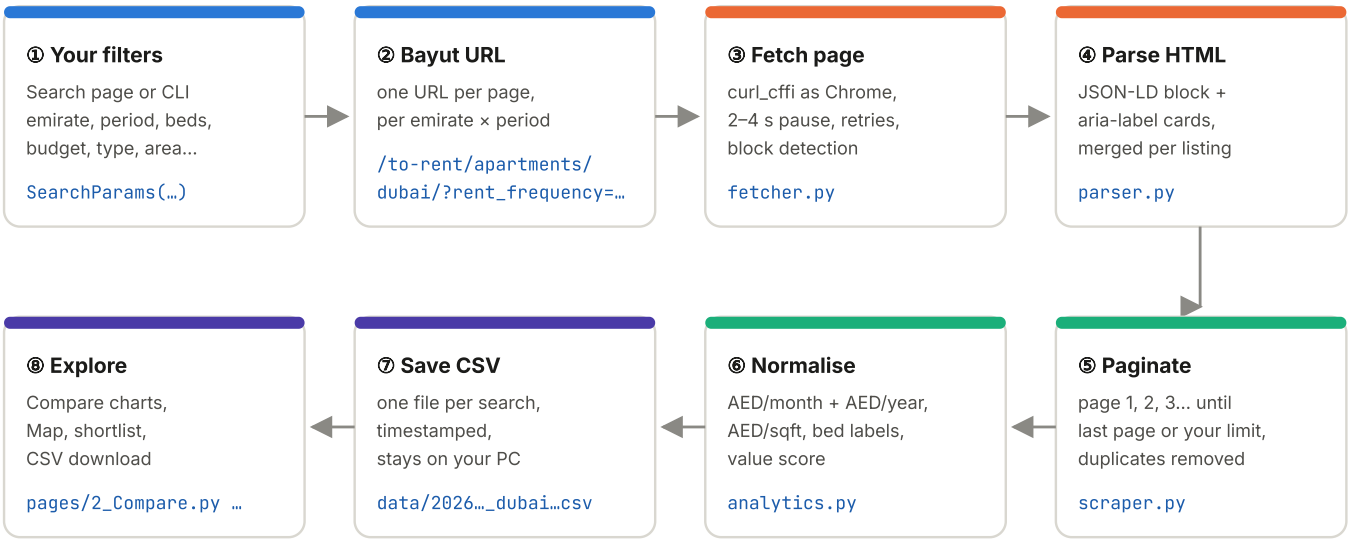


DIAGRAM The pipeline. Colours group the stages: blue = what you ask for, orange = talking to Bayut, green = cleaning, violet = keeping and using the data.

The same listing at every stage

Follow one real listing from your saved page (Bayut id **16646069**) through the pipeline:

Stage	What it looks like
① Filters	Emirate **Dubai** · Rent paid **yearly** · Property type **Any**
② URL	`https://www.bayut.com/to-rent/property/dubai/?rent_frequency=yearly`
③ Page	One full HTML page containing 24 listing cards and one JSON-LD block
④ Parsed record	`price 42000 · rent_frequency yearly · bedrooms 0 · bathrooms 1 · area_sqft 295 · latitude 25.1726 · longitude 55.3022 · location "Azizi Riviera 16, Azizi Riviera, Meydan, Dubai"`
⑥ Normalised row	**AED 3,500 / month** · **AED 42,000 / year** · AED 142.4 / sqft / year · Studio · community **Meydan** · building **Azizi Riviera 16**
⑧ In the app	Row in the results table with a **bayut ↗** link; a dot on the map 3.9 km from Burj Khalifa; counted in Meydan's median (AED 3,750 / month, 4 listings)

CHAPTER 4

Getting past Bayut's bot check

The first version of the rebuilt app returned nothing. You found the reason: Bayut checks **how** a client connects, not only what it says it is. Here is what happens and how the app handles it now.

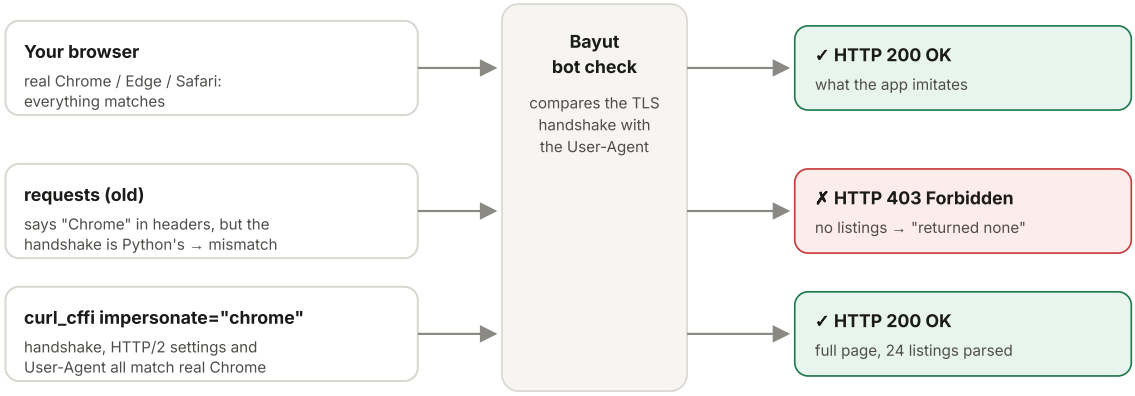


DIAGRAM Same URL, different results. The bot check looks at the connection fingerprint, so fixing headers alone never helped.

The three engines

Engine	How it fetches	Speed	When to use
chrome default	`curl_cffi` with Chrome's real TLS fingerprint. No browser window.	Fast	Always first. This is what worked in your tests.
browser	Playwright drives a headless Chromium that renders the page like a visitor.	Slow	If *chrome* also gets 403s. Needs `pip install playwright` + `playwright install chromium`.
requests	Plain Python HTTP with browser headers.	Fast	Kept for completeness; Bayut blocks it.

What the chrome engine does on every page

Wait 2–4 s — random pause since the last request (Search → Advanced → Delay)

→

GET the URL — as Chrome; only Accept and Accept-Language headers are added

→

429 / 5xx? — wait 5 s, 10 s... and retry (3 attempts)

→

403 or captcha? — new session with another fingerprint: chrome → chrome131 → edge101

→

Still blocked? — stop, keep the listings collected so far, save the page to `data/debug/`

✓ Why the app does not send its own User-Agent with curl_cffi

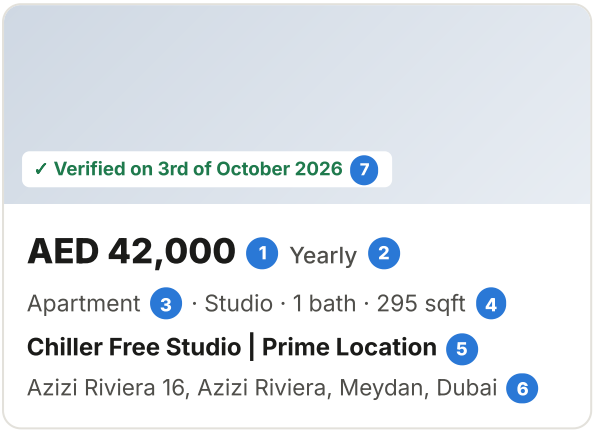
curl_cffi already sends the User-Agent that belongs to the impersonated browser. Overriding it with a different one recreates exactly the mismatch the bot check looks for. Your working snippet did the same: only `Accept` and `Accept-Language`.

CHAPTER 5

How a listing becomes data

Each results page holds every listing twice: once as the visible card and once as hidden structured data for search engines. The parser reads both and merges them, so if Bayut changes one of them the other still works.

Source A: the visible card, found by aria-labels




```
<!-- simplified from your saved page -->
<li aria-label="Listing">
 <a aria-label="Listing link" href="/property/details-
16646069.html">
 1 <span aria-label="Price">42,000</span>
 2 <span aria-label="Frequency">Yearly</span>
 3 <span aria-label="Type">Apartment</span>
 4 <span aria-label="Studio">Studio</span>  or "Beds"
   <span aria-label="Baths">1</span>
   <span aria-label="Area">295 sqft</span>
 5 <h2 aria-label="Title">Chiller Free Studio…</h2>
 6 <div aria-label="Location">Azizi Riviera 16, …, Dubai
 7 <div aria-label="Property Verified Button">…
</li>
```

The class names on these same elements (`eff033a6` , `_57d840c6` …) are randomly generated at build time, which is why the old code broke. The `aria-label` s describe meaning for screen readers and stay the same.

Source B: the hidden JSON-LD block

```
<script type="application/ld+json">  (one block per page, all 24 listings)
{ "@type": "RealEstateListing",
  "url": "https://www.bayut.com/property/details-16646069.html",
  "mainEntity": { "accommodationCategory": "Apartment",
    "geo": { "latitude": 25.17257959, "longitude": 55.30223250 },   ← GPS, no geocoding needed
    "address": { "addressRegion": "Dubai", "addressLocality": "Meydan" },
    "floorSize": { "value": "295", "unitText": "SQFT" },
    "numberOfBedrooms": 0, "numberOfBathroomsTotal": 1,
    "priceSpecification": { "price": 42000, "priceCurrency": "AED", "unitText": "yearly" },
    "realEstateAgent": { "name": "…", "memberOf": { "name": "Creative Lines Real Estate" } } } }
```

Merged result: one clean row

Card gives		JSON-LD gives		Row (listing 16646069)
full location path, building, sub-community, title, verified date, TruBroker badge	+	clean numbers, GPS, agent, agency, phone, photo URL	=	Apartment · Studio · 1 bath · 295 sqft · AED 42,000 yearly → 3,500/mo · Meydan › Azizi Riviera › Azizi Riviera 16 · 25.1726, 55.3022 · Creative Lines Real Estate · verified 3 Oct 2026

ℹ Merge rule

Cards are read first (to keep Bayut's order), then JSON-LD values are laid on top because its numbers are cleaner. An empty value never overwrites a filled one. Records are matched on the Bayut listing id from the URL (`details-16646069.html`).

CHAPTER 6

Filters → Bayut URL

Every filter you set becomes one part of the Bayut search address. Knowing which part changes helps you check a search: open the printed URL in your browser, and you'll see exactly what the app saw.

Anatomy of a search URL

```
https://www.bayut.com/to-rent/apartments/dubai/dubai-marina/page-3/?rent_frequency=monthly&furnishing_status=furnished&beds_in=1,2&price_max=9000&sort=price_asc
```

Property type · Emirate · Area (optional) · Page number (2+) · Filters after the **?** , joined with **&**

Every filter and what it changes

App control	Becomes in the URL	Notes & example values
Emirates	`/dubai/` path segment	dubai, abu-dhabi, al-ain, sharjah, ajman, ras-al-khaimah, umm-al-quwain, fujairah. **Each emirate is a separate search.**
Rent paid	`rent_frequency=monthly`	monthly or yearly in the app (weekly/daily in the CLI). Always written, even for yearly, so labels never lie. **Each period is a separate search.**
Property type	`/apartments/` path segment	Any residential → `property`, Apartment → `apartments`, Villa → `villas`, Townhouse → `townhouses`, Penthouse → `penthouse`, Hotel apartment, Villa compound
Area (optional)	`/dubai-marina/` after the emirate	Bayut's area slug as it appears in the URL. Typing "Dubai Marina" is converted to `dubai-marina`. Applies to the **first** emirate only.
Bedrooms	`beds_in=0,1,2`	Studio = 0. Several values are joined with commas. Empty = any.
Furnishing	`furnishing_status=furnished`	Left out for "all". Bayut cards don't show furnishing, so the CSV column records what you searched for.
Min / Max budget AED per month	`price_min=…&price_max=…`	Converted to the search's period: monthly searches use the number as is, **yearly searches multiply by 12**. 0 = no limit.
Min / Max size	`area_min=…&area_max=…`	In sqft. Under *Advanced*.
Sort	`sort=price_asc`	Popular (nothing added), Newest `date_desc`, Lowest price `price_asc` (app default), Highest `price_desc`, Verified first `verified_score`.
Pages per search	`/page-2/`, `/page-3/`…	Page 1 has no page segment. Stops early when Bayut runs out of results.

Engine and Delay don't change the URL, only how pages are downloaded (chapter 4). Baths is available on the command line only: `--baths 2 3` → `baths_in=2,3`.

Watch a URL grow, one filter at a time

These are the exact URLs the app builds. Each row adds one filter to the one above (generated by `build_search_url`, not typed by hand):

Starting point	`https://www.bayut.com/to-rent/property/dubai/?rent_frequency=yearly`
Rent paid: monthly	`…/to-rent/property/dubai/?rent_frequency=monthly`
Type: Apartment	`…/to-rent/apartments/dubai/?rent_frequency=monthly`
Area: Dubai Marina	`…/to-rent/apartments/dubai/dubai-marina/?rent_frequency=monthly`
Bedrooms: 1 BR, 2 BR	`…/dubai-marina/?rent_frequency=monthly&beds_in=1,2`
Max budget: 9,000 / month	`…/dubai-marina/?rent_frequency=monthly&beds_in=1,2&price_max=9000`
Furnished + Lowest price	`…/?rent_frequency=monthly&furnishing_status=furnished&beds_in=1,2&price_max=9000&sort=price_asc`
...and its 3rd page	`…/dubai-marina/page-3/?rent_frequency=monthly&…&sort=price_asc`

One click can mean several searches

Emirates and rent periods multiply. With the default selection the app runs **4 searches**, each with its own URL and its own pages:

	monthly	yearly
Dubai	search 1 budget 9,000 → `price_max=9000`	search 2 budget 9,000 → `price_max=108000`
Abu Dhabi	search 3	search 4

4 searches × 5 pages = up to **20 pages ≈ 480 listings**. The URLs are listed under "4 search URL(s)" right after you click Search.

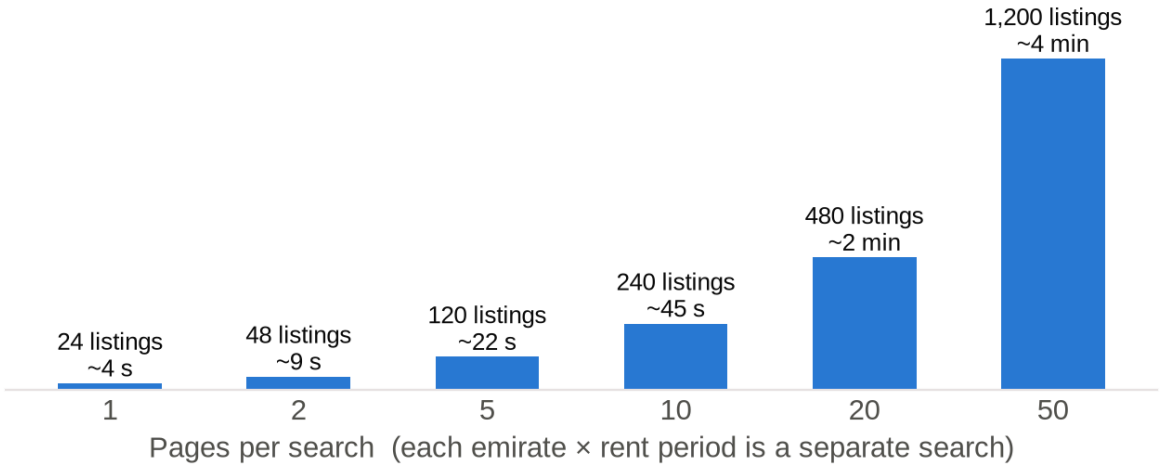
✗ The old URL bug, and why a builder matters

The old `get_url` added each filter with its own `?`:

```
…/dubai/?rent_frequency=monthly?furnishing_status=furnished
```

A URL can have only one `?`, so Bayut read the first filter as `monthly?furnishing_status=furnished`. Both filters were quietly lost. The new builder encodes all filters together with `&`.

How many pages should I scrape?



ESTIMATE Listings collected per search, and rough time at the default 2–4 s delay plus download and parse time. Multiply by the number of searches.

Rules of thumb

Your goal	Pages	Sort	Why
Find a flat now	2–5	Lowest price	The cheapest matches come first; you only need the top of the list.
Compare a few areas	10–20	Popular	Enough listings per community (3+) for a trustworthy median, without a cheap-end bias.
Monthly vs yearly premium	10+ per period	Popular	Both periods must have 2+ listings in the same community and bedroom count.
Whole-emirate overview	30–50	Popular	Slow (several minutes per search). Keep the delay, run it while you do something else.
One specific building or area	1–3	any	Use the Area field; a narrow search has few pages anyway, and the app stops at the last one.

ℹ The app never fetches more pages than exist

Page 1 tells the app the total (e.g. "1 to 24 of 118,858 Properties"). It caps the page count at that total ÷ 24, stops as soon as a page brings no new listings, and skips listings it has already seen.

CHAPTER 7

Three worked searches

From a real-life goal to the settings to type, the URLs the app sends, and where to look in the results.

A · "A furnished 1–2 bed in Dubai Marina I can pay monthly, up to AED 9,000"

Emirates	**Dubai** only
Rent paid	**monthly**
Property type	Apartment
Bedrooms	1 BR, 2 BR
Furnishing	furnished
Area	`dubai-marina`
Max budget	9000
Pages	5

1 search → URL:

```
…/to-rent/apartments/dubai/dubai-marina/?rent_frequency=monthly&furnishing_status=furnished&beds_in=1,2&price_max=9000&sort=price_asc
```

Then: Compare → *Shortlist* for listings below the Marina median for the same bedroom count; Map → "Dubai Marina" or "Dubai Internet City / Media City" to check distance to work.

B · "Is a monthly listing worth it, or should I sign a yearly contract?"

Emirates	**Dubai + Abu Dhabi**
Rent paid	**monthly + yearly**
Property type	Apartment
Bedrooms	1 BR
Max budget	9000 (per month)
Pages	10

4 searches, e.g. Abu Dhabi yearly:

```
…/to-rent/apartments/abu-dhabi/?rent_frequency=yearly&beds_in=1&price_max=108000
```

Note the budget became **108,000** for yearly (9,000 × 12). **Then:** Compare → *Paying monthly vs a yearly contract* shows the premium per community (chapter 10).

C · "A 3–4 bed villa in Khalifa City, Abu Dhabi, at least 2,500 sqft"

Emirates	**Abu Dhabi**
Rent paid	yearly
Property type	Villa
Bedrooms	3 BR, 4 BR
Area	`khalifa-city`
Min size	2500 (Advanced)

1 search → URL:

```
…/to-rent/villas/abu-dhabi/khalifa-city/?rent_frequency=yearly&beds_in=3,4&area_min=2500
```

Then: Compare → *Where is it cheaper?* → Rank by **Cheapest per sqft**: for villas, price per sqft says more than price alone.

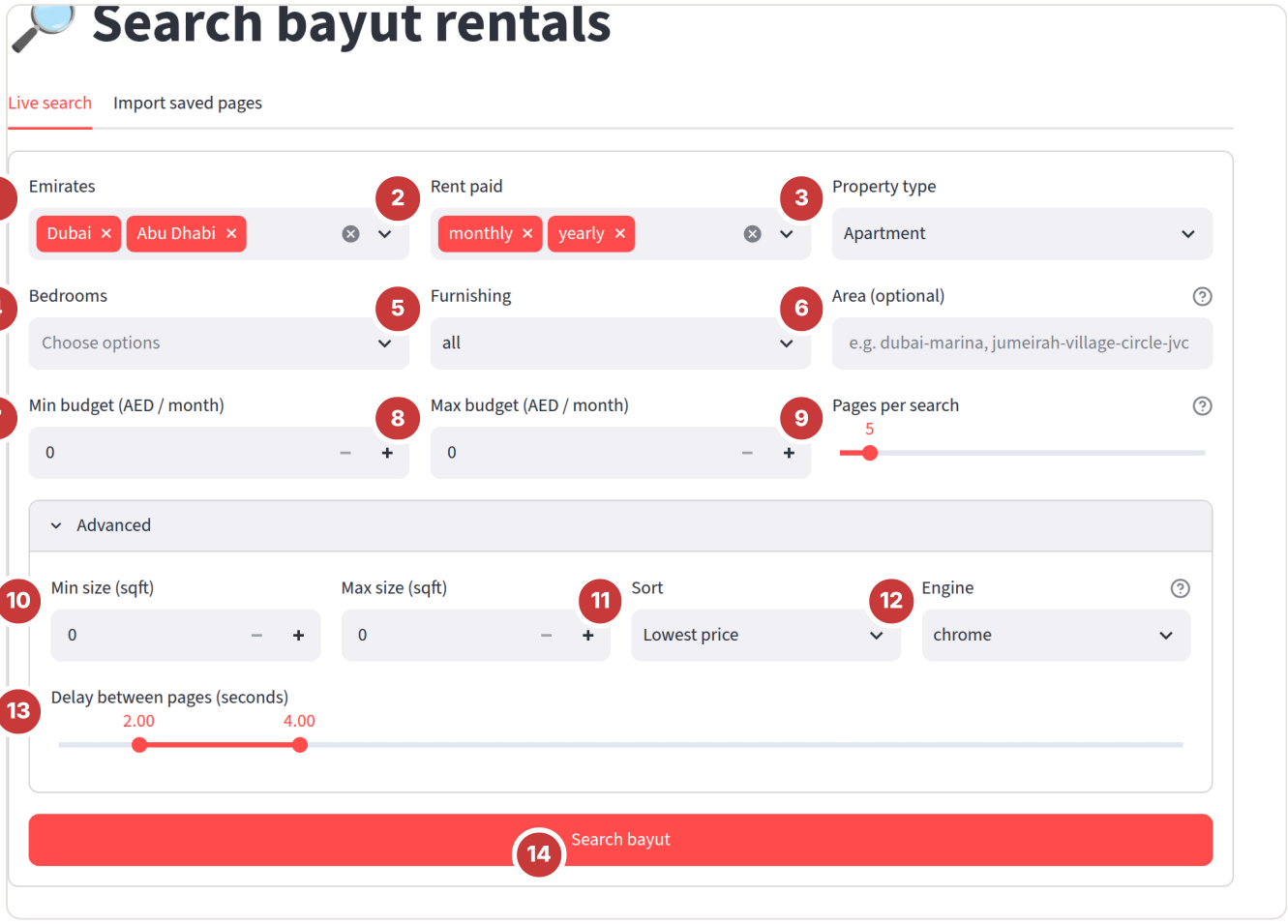
✓ Find the right area slug in 10 seconds

Open bayut.com, pick the area in the search box, and copy the part of the address after the emirate. For example, `bayut.com/to-rent/property/dubai/jumeirah-village-circle-jvc/` → type `jumeirah-village-circle-jvc`. Sub-areas work too: `al-quoz/al-quoz-4`.

CHAPTER 8

The Search page, control by control

Fill the form top to bottom, click the red button, and watch the progress bar. When it finishes, the results are already saved.



1. **Emirates**: one search per emirate. Default Dubai + Abu Dhabi.
2. **Rent paid**: monthly, yearly or both (both = compare later).
3. **Property type**: changes the URL path (apartments, villas...).
4. **Bedrooms**: Studio...7 BR, several allowed. Empty = any.
5. **Furnishing**: all / furnished / unfurnished.
6. **Area**: optional slug, first emirate only.
7. **Min budget**: AED per month; converted ×12 for yearly searches.
8. **Max budget**: same rule. 0 = no limit.
9. **Pages per search**: 1–50, 24 listings each.
10. **Min / Max size**: sqft, inside *Advanced*.
11. **Sort**: default "Lowest price", so the first pages hold the cheapest.
12. **Engine**: keep **chrome** (chapter 4).
13. **Delay**: random pause between pages, 2–4 s default.
14. **Search bayut**: runs every search, saves one CSV.

> ⚠ **"Lowest price" sort + few pages = a cheap-end sample**
>
> With the default sort, 5 pages give you the **cheapest** ~120 listings, not a random mix. That's ideal for finding a flat, but it pulls medians down. For fair area medians, use **Popular** sort or scrape more pages.

What you see after clicking Search

Search URLs
an expander lists every URL, so you can open one in your browser to check it

→

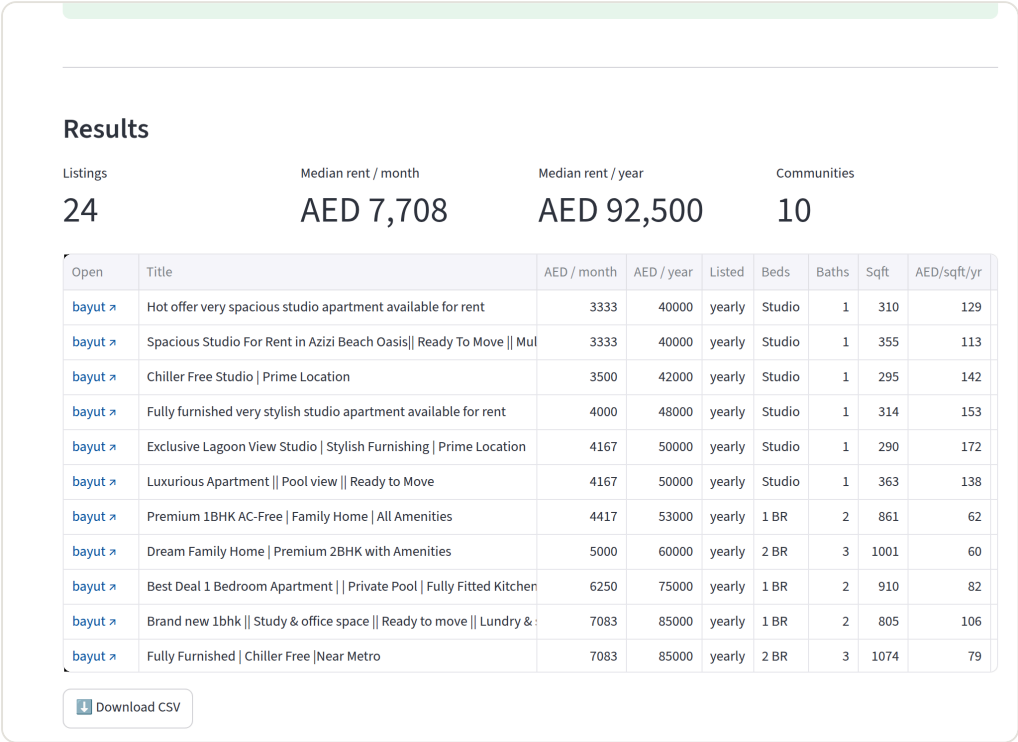
Progress bar
"dubai_apartments_monthly - page 3/5 - 72 listings"

→

Warnings
(if any) per search, e.g. blocked after page 4: earlier pages are kept

→

Green banner
"Collected N listings → saved to data/...csv"



Results

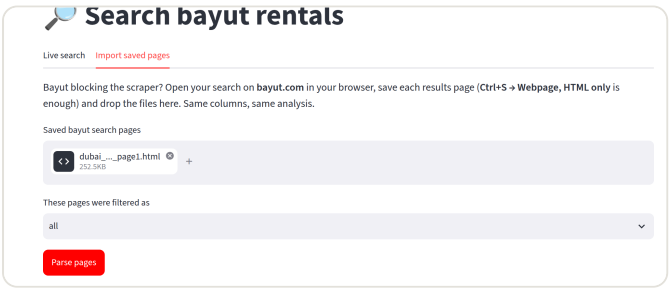
Listings	Median rent / month	Median rent / year	Communities
24	AED 7,708	AED 92,500	10

Open	Title	AED / month	AED / year	Listed	Beds	Baths	Sqft	AED/sqft/yr
bayut ↗	Hot offer very spacious studio apartment available for rent	3333	40000	yearly	Studio	1	310	129
bayut ↗	Spacious Studio For Rent in Azizi Beach Oasis|| Ready To Move || Mul	3333	40000	yearly	Studio	1	355	113
bayut ↗	Chiller Free Studio | Prime Location	3500	42000	yearly	Studio	1	295	142
bayut ↗	Fully furnished very stylish studio apartment available for rent	4000	48000	yearly	Studio	1	314	153
bayut ↗	Exclusive Lagoon View Studio | Stylish Furnishing | Prime Location	4167	50000	yearly	Studio	1	290	172
bayut ↗	Luxurious Apartment || Pool view || Ready to Move	4167	50000	yearly	Studio	1	363	138
bayut ↗	Premium 1BHK AC-Free | Family Home | All Amenities	4417	53000	yearly	1 BR	2	861	62
bayut ↗	Dream Family Home | Premium 2BHK with Amenities	5000	60000	yearly	2 BR	3	1001	60
bayut ↗	Best Deal 1 Bedroom Apartment | | Private Pool | Fully Fitted Kitchen	6250	75000	yearly	1 BR	2	910	82
bayut ↗	Brand new 1bhk || Study & office space || Ready to move || Lundry &	7083	85000	yearly	1 BR	2	805	106
bayut ↗	Fully Furnished | Chiller Free |Near Metro	7083	85000	yearly	2 BR	3	1074	79

⬇️ Download CSV

REAL DATA Results from your saved Dubai page: 24 listings, median AED 7,708 / month, 10 communities. Click a column header to sort; **bayut ↗** opens the listing; **Download CSV** exports what you see.

Plan B: import pages saved from your browser



REAL DATA The Import tab after parsing your saved page.

1. **Search on bayut.com as usual**
   Set your filters in the browser.

2. **Save each results page**
   Ctrl+S → "Webpage, HTML only" is enough. Click "next page", save again.

3. **Drop the files into "Import saved pages"**
   Tell the app how the pages were filtered (furnishing), click **Parse pages**. Same columns, same Compare and Map.

CHAPTER 9

From price to comparable numbers

Bayut shows each listing in its own rent period. The app converts every listing to both a monthly and a yearly figure, so a flat listed at "AED 6,600 Monthly" and one at "AED 66,000 Yearly" can sit in the same chart.

The conversion rules

Listed as	Multiplier to a year	AED / year	AED / month	Example
Yearly	× 1	price	price ÷ 12	42,000 → **3,500 / mo** (your listing 16646069)
Monthly	× 12	price × 12	price	6,600 → **79,200 / yr**
Weekly	× 52	price × 52	year ÷ 12	1,500 → 78,000 / yr → 6,500 / mo
Daily	× 365	price × 365	year ÷ 12	250 → 91,250 / yr → 7,604 / mo

Also computed for every listing: `yearly_per_sqft` = AED / year ÷ size (42,000 ÷ 295 = **142.4**); `bedrooms_label` = Studio, 1 BR … 7+ BR; sizes in sqm are converted to sqft (× 10.764).

Monthly listing vs yearly contract

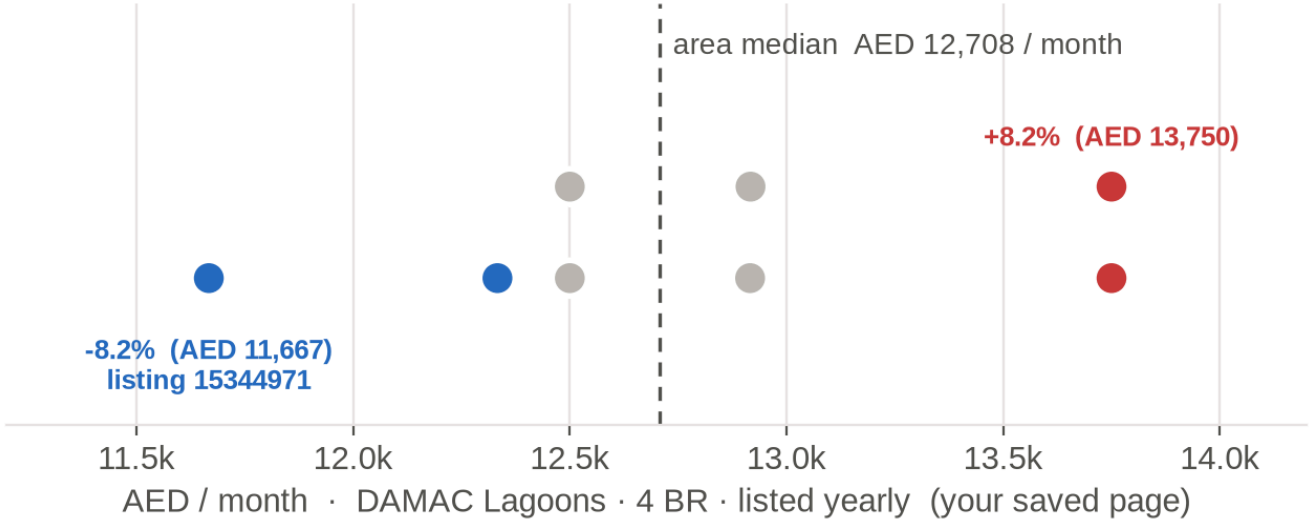


ILLUSTRATION The same type of flat costs more over 12 months when paid monthly. The Compare page measures this premium per community and bedroom count from your own data.

> **⚠ A monthly listing is often a different product**
>
> Monthly listings are frequently furnished, flexible, and sometimes include bills or chiller. Part of the premium pays for that. Compare like with like: filter Furnishing, and read the listing before deciding.

The value score: "vs area median"

For every listing, the app finds **comparable listings**: same emirate, same community, same bedroom count and same rent period. It then shows how far the listing is from their median, as a percentage. **Negative = cheaper than similar listings.** A group needs at least 3 listings, otherwise the score is left blank.



REAL DATA The 8 four-bed DAMAC Lagoons listings on your saved page. Median AED 12,708 / month; listing 15344971 at AED 11,667 is **−8.2%**, the best value in its group; two at 13,750 are +8.2%. Blue = more than 2% below median, red = more than 2% above, grey = within ±2%.

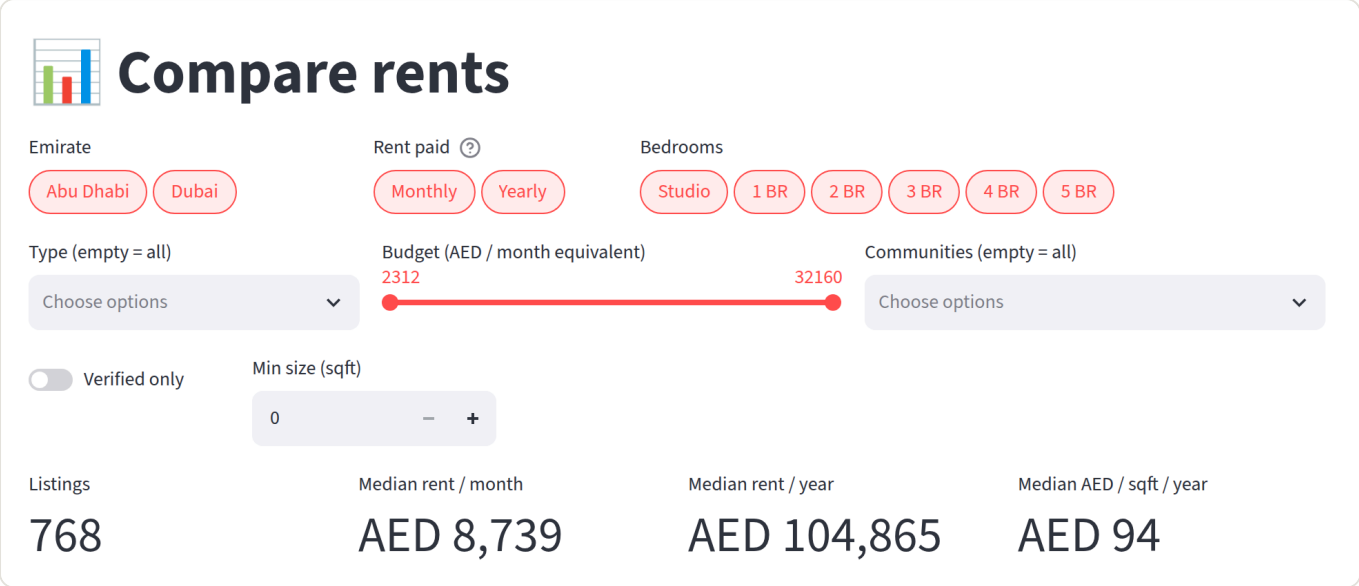
Formula: vs_area_median = (listing AED/month ÷ group median − 1) × 100 → (11,667 ÷ 12,708 − 1) × 100 = −8.2%.

CHAPTER 10

The Compare page

One page, six blocks, read from top to bottom. The filters at the top apply to everything below them. The screenshots here use SAMPLE DATA covering both emirates and both rent periods, so every block is filled.

A · Data, filters and headline numbers



SAMPLE DATA Pill buttons toggle emirates, rent periods and bedroom counts. Turning all pills in a group off means "no filter". The budget slider is always **per month**, even for yearly listings.

Pick the data in the sidebar

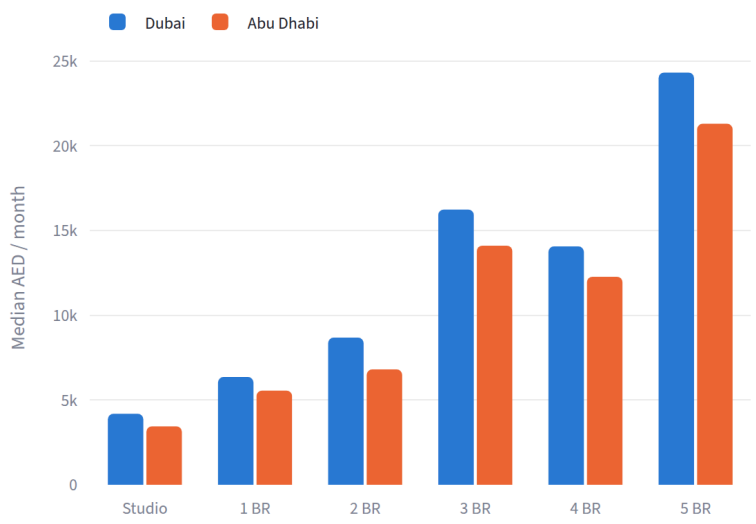
"Datasets" lists the latest search plus every CSV in `data/`. Select several to combine them, e.g. last week's monthly scrape + today's yearly scrape. If the same listing appears twice, the newer file wins.

Read the 4 numbers

Listings in view, **median** rent per month and per year, and median AED per sqft per year. The median is the middle listing, so one luxury penthouse can't distort it the way it would an average.

B · Typical rent by bedrooms

Typical rent by bedrooms



Median AED / month (× 12 for the yearly equivalent)

Bedrooms	Abu Dhabi	Dubai
Studio	3,449	4,184
1 BR	5,555	6,356
2 BR	6,805	8,679
3 BR	14,106	16,235
4 BR	12,268	14,060
5 BR	21,291	24,316

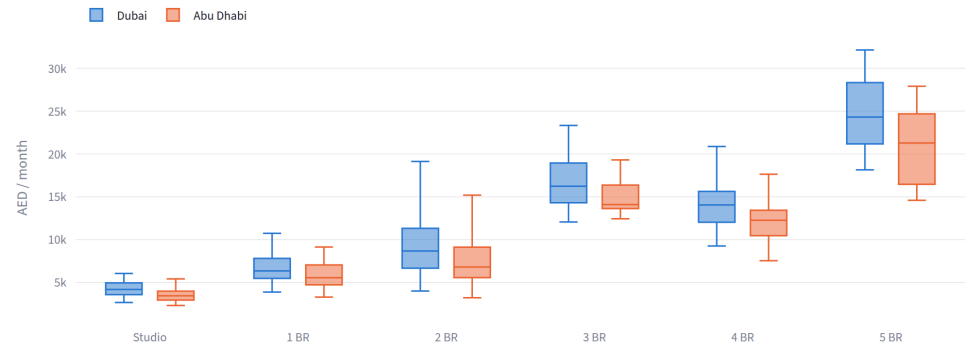
SAMPLE DATA Median AED / month per bedroom count, Dubai (blue) next to Abu Dhabi (orange). Hover a bar for the yearly median and the listing count; the table on the right gives the exact numbers.

✓ How to use it

Find your bedroom count and read across: in this sample, a 2 BR is AED 8,679 / month in Dubai vs 6,805 in Abu Dhabi. Then check whether moving up one bedroom costs more than you'd expect. A big jump means the next size up is comparatively expensive.

C · Price spread

Price spread



Box = middle 50% of listings, line = median. Very long whiskers usually mean a mix of budget and luxury buildings - narrow down by community.

SAMPLE DATA Box = the middle half of listings; the line inside = median; whiskers = the normal range.

Short box

Prices are consistent; the median is a reliable target for negotiating.

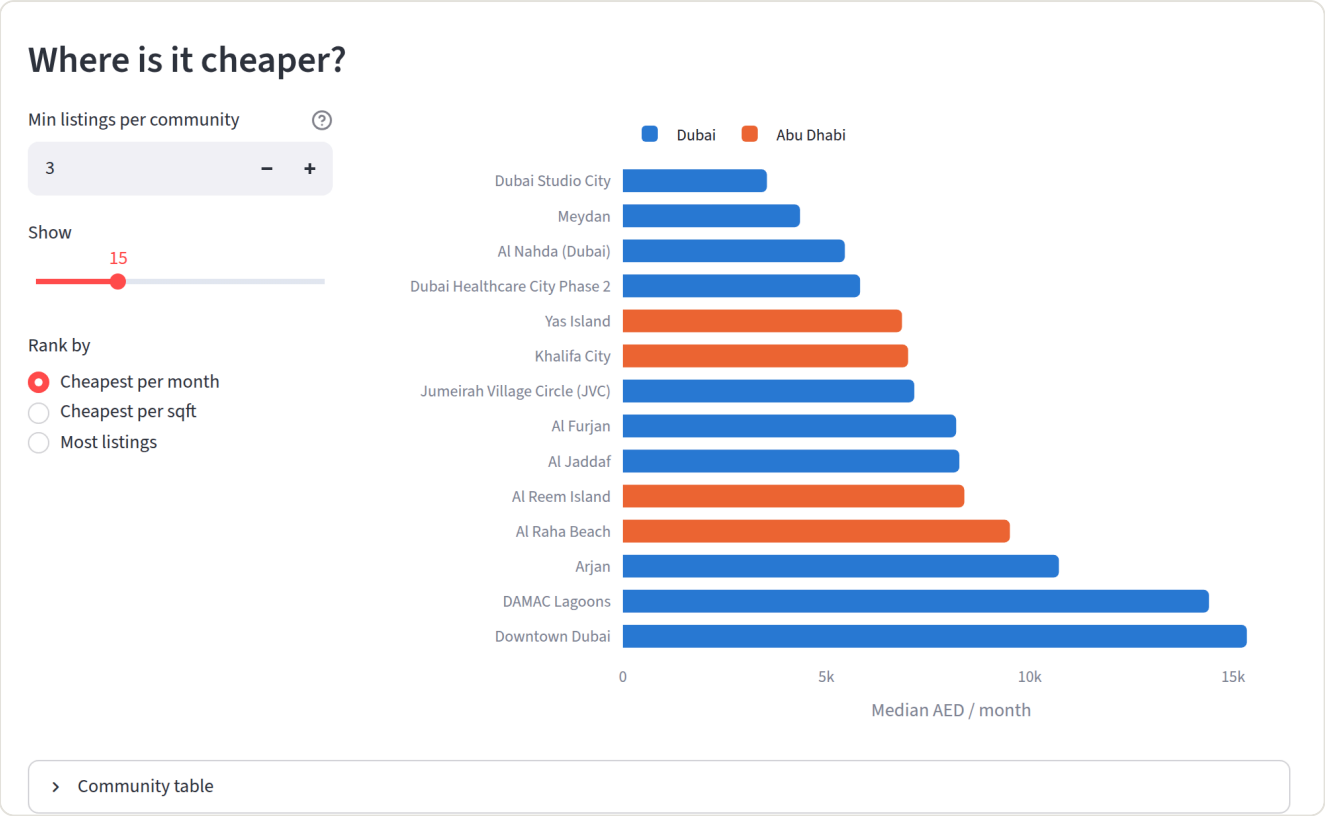
Tall box / long whiskers

A mix of budget and premium buildings. Narrow it down with the **Communities** filter.

Boxes overlap

Two emirates or sizes cost about the same. Choose on location and quality, not price.

D · Where is it cheaper?



SAMPLE DATA Communities ranked by median AED / month, coloured by emirate. Controls on the left: minimum listings per community (to ignore thin data), how many to show, and the ranking.

Rank by	Answers	Best for
Cheapest per month	"Where can I pay the least?"	Fixed budget, any size
Cheapest per sqft	"Where do I get the most space per dirham?"	Villas, families, comparing sizes
Most listings	"Where is there the most choice?"	Negotiating power, quick moves

Open **Community table** under the chart for every community with listings, median monthly, yearly, minimum, median size and AED/sqft.

E · Paying monthly vs a yearly contract

Paying monthly vs a yearly contract

Across **34** comparable community × bedroom groups, monthly listings cost **+21%** over a full year compared to yearly contracts.

Emirate	Community	Beds	Yearly contract (AED/yr)	Monthly × 12 (AED/yr)	Monthly premium	# yearly	# monthly
Dubai	Al Nahda (Dubai)	1 BR	63920	60420	-6%	8	8
Abu Dhabi	Yas Island	2 BR	95010	90840	-4%	10	9
Abu Dhabi	Al Reem Island	1 BR	68535	66480	-3%	6	7
Abu Dhabi	Khalifa City	Studio	41380	41520	+0%	13	17
Dubai	Downtown Dubai	2 BR	181280	184020	+2%	8	8
Abu Dhabi	Al Raha Beach	4 BR	148480	156720	+6%	13	19
Dubai	Arjan	3 BR	187775	202440	+8%	8	8
Dubai	Dubai Studio City	Studio	41245	44460	+8%	8	8
Abu Dhabi	Yas Island	1 BR	69830	75840	+9%	9	3
Abu Dhabi	Yas Island	Studio	39290	43080	+10%	17	15

SAMPLE DATA For each community and bedroom count with at least 2 monthly *and* 2 yearly listings: the yearly-contract median next to the monthly median × 12.

Reading a row

Abu Dhabi · Yas Island · Studio: yearly contract 39,290 / yr; monthly listings 43,080 / yr → **+10%**. Paying monthly costs about AED 3,800 more over a year, which buys flexibility (and often furniture).

Negative premium?

Rows like **−6%** mean monthly listings were *cheaper* in that group: usually few listings, different buildings, or the cheap-end sort. Check the **# yearly / # monthly** counts before trusting a row.

ℹ Empty block?

This block needs both rent periods. Run a search with **Rent paid = monthly + yearly** and enough pages (10+), or combine a monthly and a yearly dataset in the sidebar.

F · Shortlist: priced below similar listings

Shortlist: priced below similar listings

Open	Title	AED / month	AED / year	Listed	Beds	Baths	Sqft	AED/sqft/yr
bayut ↗	Dream Family Home | Premium 2BHK with Amenities	3237	38840	yearly	2 BR	3	1001	39
bayut ↗	Dream Family Home | Premium 2BHK with Amenities	3216	38590	yearly	2 BR	3	1001	39
bayut ↗	Dream Family Home | Premium 2BHK with Amenities	3800	45600	monthly	2 BR	3	1001	46
bayut ↗	Dream Family Home | Premium 2BHK with Amenities	4220	50640	monthly	2 BR	3	1001	51
bayut ↗	Fully Furnished | Chiller Free |Near Metro	4632	55590	yearly	2 BR	3	1074	52
bayut ↗	Dream Family Home | Premium 2BHK with Amenities	3263	39160	yearly	2 BR	3	1001	39
bayut ↗	Dream Family Home | Premium 2BHK with Amenities	3268	39210	yearly	2 BR	3	1001	39
bayut ↗	Dream Family Home | Premium 2BHK with Amenities	4760	57120	monthly	2 BR	3	1001	57
bayut ↗	Premium 1BHK AC-Free | Family Home | All Amenities	3740	44880	monthly	1 BR	2	861	52
bayut ↗	Dream Family Home | Premium 2BHK with Amenities	4780	57360	monthly	2 BR	3	1001	57
bayut ↗	Dream Family Home | Premium 2BHK with Amenities	5580	66960	monthly	2 BR	3	1001	67

⬇️ Download CSV

SAMPLE DATA The 50 listings with the most negative value score (chapter 9), cheapest relative to their peers first. The **vs area median** column is further right in the app (scroll the table sideways).

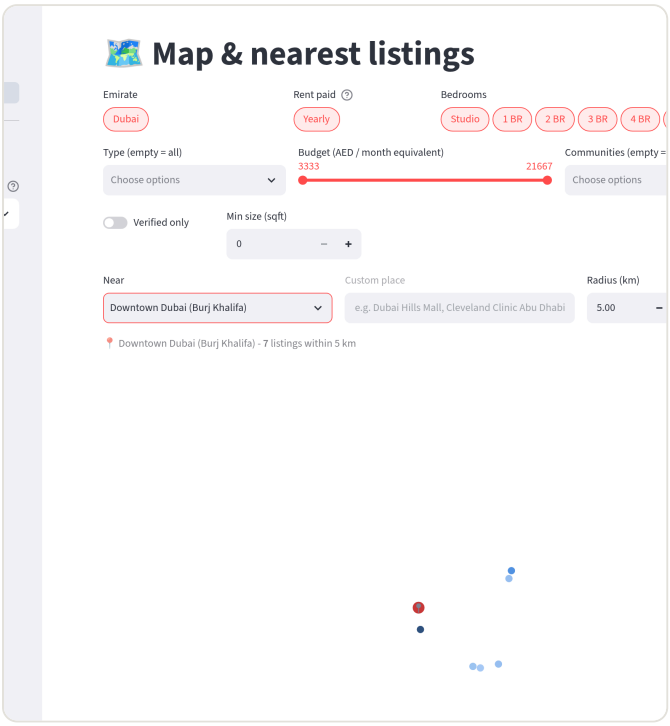
✓ A good routine

1. Filter to your bedrooms and budget at the top. 2. Read the shortlist. 3. Open the top 5–10 with **bayut ↗**. 4. Check photos, floor, view, cheques and fees, since a low price sometimes has a reason. 5. **Download CSV** to keep your shortlist.

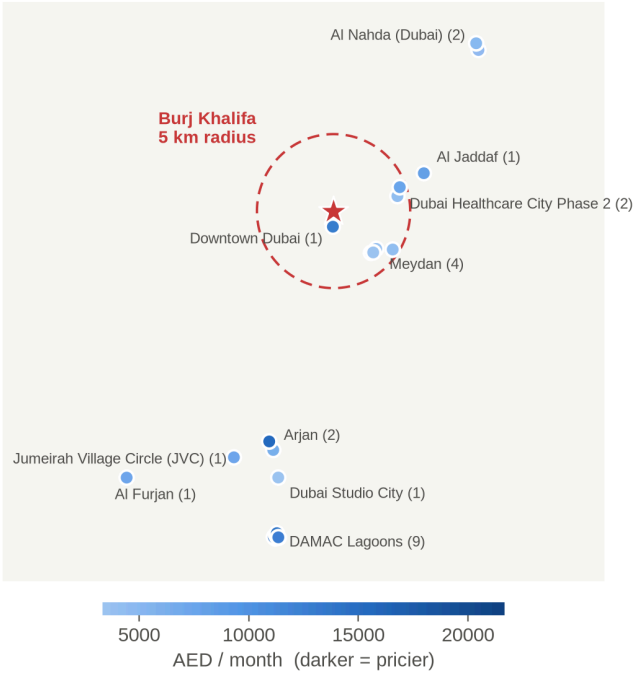
CHAPTER 11

The Map page

Same filters as Compare, plus a reference point. Every listing has exact GPS from Bayut, so distances are measured from the listing itself, not from an area centre.



REAL DATA Your saved page with "Downtown Dubai (Burj Khalifa)" selected: 7 listings within 5 km. The red marker is the landmark. On your PC, the dots sit on an OpenStreetMap street map; this capture tool couldn't load the map tiles.



REAL DATA The same 24 listings drawn as an illustration: darker = pricier, dashed circle = 5 km around Burj Khalifa, (n) = listings per community.

How distance is measured

1 · Listing GPS
Latitude and longitude come from Bayut's own data for each listing, usually the building's position.

2 · Your point
A built-in landmark, or a place you type, looked up on OpenStreetMap.

3 · Straight line
Great-circle distance in km (haversine formula). It's a first filter, not a drive time: check commutes in a maps app.

Controls

Control	What it does
Near	16 built-in landmarks, no internet lookup needed: Downtown Dubai, DIFC, Business Bay, Dubai Marina, Internet/Media City, Al Barsha / MoE, TECOM, Silicon Oasis, DXB, Deira, Abu Dhabi Corniche, Al Reem Island, Saadiyat, Yas Island, Khalifa City, AUH.
Custom place…	Type any place, e.g. *Dubai Hills Mall* or *Cleveland Clinic Abu Dhabi*. It's looked up on OpenStreetMap (free, UAE only) and cached for a day.
Radius (km)	The map shows only listings within this straight-line distance.
Nearest	How many of the closest listings go in the table below the map, sorted by distance.



Nearest 20 to Downtown Dubai (Burj Khalifa)

km away	Open	Title	AED / month	AED / year	List
1.0	bayut ↗	Burj Khalifa View | Furnished | Prime Location	12917	155000	yea
3.6	bayut ↗	Fully furnished very stylish studio apartment available for rent	4000	48000	yea
3.6	bayut ↗	Hot offer very spacious studio apartment available for rent	3333	40000	yea
3.9	bayut ↗	Chiller Free Studio | Prime Location	3500	42000	yea
4.3	bayut ↗	Luxurious Apartment || Pool view || Ready to Move	4167	50000	yea
4.4	bayut ↗	Exclusive Lagoon View Studio | Stylish Furnishing | Prime Location	4167	50000	yea
4.5	bayut ↗	Elegant 2BR Apartment | Binghatti Avenue | Prime Dubai Location	7500	90000	yea
6.4	bayut ↗	Chiller free || Brand New building || 2bhk apartment || with all ameni	7917	95000	yea
14.2	bayut ↗	Premium 1BHK AC-Free | Family Home | All Amenities	4417	53000	yea

REAL DATA "Nearest 20 to Downtown Dubai": the closest is a 2 BR at 1.0 km (AED 12,917 / month); a studio 3.6 km away costs AED 3,333.

✓ Typing a custom place

Be specific and add the area: "Mall of the Emirates, Al Barsha" works better than "MOE". If nothing is found, try a nearby landmark from the list.

ℹ Several listings, one pin

Listings in the same building share coordinates, so their dots overlap. Hover the pin, or use the table, which lists every one.

CHAPTER 12

Command line

Everything the Search page does is also available as a command, handy for repeating the same search every few days or scheduling it.

Scrape: 1–2 bed apartments in Dubai, listed monthly, up to AED 9,000, 5 pages

```
> python -m bayut_finder scrape --emirate dubai --frequency monthly --category apartments --beds 1
2 --price-max 9000 --pages 5
  dubai_apartments_monthly_beds-1-2: page 1/5 - 24 listings
  dubai_apartments_monthly_beds-1-2: page 2/5 - 48 listings
  …
```

Scrape both emirates and both periods into one CSV

```
> python -m bayut_finder scrape --emirate dubai abu-dhabi --frequency monthly yearly --pages 3
```

Parse pages you saved from your browser (real output from your saved page)

```
> python -m bayut_finder parse bayut_page.html
24 listings
emirate  rent_frequency  bedrooms_label
Dubai    yearly          1 BR               6250.0
                         2 BR               7500.0
                         3 BR              15417.0
                         4 BR              12708.0
                         5 BR              21667.0
                         Studio             3750.0

Cheapest communities (median AED / month):
emirate     community  listings  median_monthly  median_yearly
  Dubai        Meydan         4          3750.0        45000.0
  Dubai DAMAC Lagoons         9         12916.0       154990.0

Saved 24 listings → data\20261017-0846_saved-pages.csv
```

All options

Option	Values	Same as in the app
`--emirate`	one or more of dubai, abu-dhabi, al-ain, sharjah, ajman, ras-al-khaimah, umm-al-quwain, fujairah	Emirates
`--frequency`	yearly, monthly, weekly, daily (one or more)	Rent paid
`--category`	property, apartments, villas, townhouses, penthouse, hotel-apartments, villa-compound	Property type
`--location`	area slug, e.g. `al-reem-island`	Area
`--furnishing`	all, furnished, unfurnished	Furnishing
`--beds` / `--baths`	numbers, 0 = studio	Bedrooms / (CLI only)
`--price-min` / `--price-max`	AED **in the chosen period**	Budget, **but not converted** (see ⚠ below)
`--area-min` / `--area-max`	sqft	Min / Max size
`--sort`	date_desc, price_asc, price_desc, verified_score	Sort
`--pages`, `--engine`, `--delay MIN MAX`, `--out`	pages per search (3), chrome / browser / requests, seconds, CSV path	Pages, Engine, Delay

✗ CLI budget is per period, not per month

In the app, the budget is per month and converted for yearly searches. On the command line, `--price-max` is used **as is** for every period. So `--frequency monthly yearly --price-max 9000` caps yearly listings at AED 9,000 a year, which finds nothing. Run monthly and yearly as two commands with their own budgets (9000 and 108000). Fixing this is idea #1 in chapter 16.

CHAPTER 13

Your data files

```
bayutfinder/
├─ Home.py                ← streamlit run this
├─ pages/                 ← Search, Compare, Map
├─ bayut_finder/          ← the engine
├─ tests/                 ← pytest, saved page
└─ data/                  ← created on first
search
   ├─ 20261017-0912_dubai-monthly+
   │  dubai-yearly_apartments.csv
   ├─ 20261017-0930_imported-pages.csv
   └─ debug/              ← only when a page
fails
      └─ dubai_property_monthly_page1.html
```

- **One CSV per search click**, named *date-time_what-you-searched.csv*, so nothing is overwritten and you build a history.
- The **Compare** and **Map** sidebars list these files, newest first.
- `data/` is in `.gitignore`: your scraped data never goes to GitHub.
- **debug/** holds the raw page Bayut sent when a search produced nothing. Open it in a browser to see why (chapter 15).
- Delete old CSVs whenever you like; the app only reads what's there.

Every column, with a real example (listing 16646069)

Column	Example	Meaning
`listing_id`	16646069	Bayut's id from the URL; used to remove duplicates
`title`	Chiller Free Studio | Prime Location	Agent's headline
`property_type`	Apartment	Apartment, Villa, Townhouse, Penthouse...
`bedrooms` / `bedrooms_label`	0 / Studio	Number (0 = studio) and display label
`bathrooms`, `area_sqft`, `price`, `rent_frequency`	1, 295, 42000, yearly	Size always in sqft; price and period exactly as listed on Bayut
`rent_monthly`, `rent_yearly`, `yearly_per_sqft`	3500, 42000, 142.4	Normalised rent (chapter 9) and AED per sqft per year
`emirate`, `community`, `sub_community`, `building`, `location`	Dubai, Meydan, Azizi Riviera, Azizi Riviera 16 (+ the full line)	Location levels split from the card's full location line; emirate + community drive groups, medians and colours
`latitude`, `longitude`	25.17258, 55.30223	GPS from the JSON-LD block
`furnishing`	Any	What you *searched* for (Furnished / Unfurnished / Any)
`verified`, `verified_on`, `trubroker`	True, 3rd of October 2026, True	Bayut's "Verified" badge and date; agent has the TruBroker badge
`agent`, `agent_phone`, `agency`	..., +9715..., Creative Lines Real Estate	Contact details shown on the listing
`url`, `image`, `search_*`, `scraped_at`	.../property/details-16646069.html, photo URL, dubai, yearly, 1, 2026-10-17T...	Listing link and first photo; which search and page produced the row, and when

CHAPTER 14

Things to be careful of

The app gives you good numbers fast, but not perfect ones. These are the 15 things most likely to mislead you, roughly in order of importance.

1 · Filter names are inferred HIGH

Bayut doesn't publish its URL parameters. `beds_in`, `price_max`, `furnishing_status` and `sort` follow Bayut's site, but if Bayut renames one, it's silently ignored. **Check:** open a printed URL and confirm bayut.com shows the filter applied.

2 · CLI budget isn't converted HIGH

`--price-max` is per chosen period. With mixed periods it's wrong for one of them (chapter 12). The app converts correctly.

3 · Area applies to the first emirate only MEDIUM

Type `dubai-marina` with Dubai + Abu Dhabi selected: Dubai is limited to the Marina, Abu Dhabi is searched in full. Put the emirate the area belongs to first, or search it alone.

4 · Small samples MEDIUM

A median of 3 listings is a hint, not a fact. Scrape 10+ pages, raise "Min listings per community", and check counts in the tables.

5 · Sort order skews medians MEDIUM

"Lowest price" + few pages = the cheapest slice. Great for hunting, biased for market medians. Use "Popular" when measuring an area.

6 · Monthly ≠ yearly product MEDIUM

Monthly listings are often furnished, flexible, and may include bills. The premium isn't all markup; compare with the Furnishing filter set.

7 · Asking price, not final price MEDIUM

Listings show what landlords ask. Yearly rents are often negotiable and depend on the number of cheques (fewer cheques usually means a better price).

8 · Hidden costs aren't in the price MEDIUM

Typical extras: agency fee, security deposit, Ejari (Dubai) / Tawtheeq (Abu Dhabi) registration, DEWA / ADDC utilities, district cooling (chiller), moving. Budget for the total, and confirm current rules with the agent.

9 · Duplicate listings LOW

The same flat can be posted by several agencies under different ids. Counts can run high; the value score can compare a flat with itself. Look for same building + size + price.

10 · Furnishing column = your filter LOW

Cards don't show furnishing, so the column records the search filter ("Any" for all), not the flat itself.

11 · Distances are straight lines LOW

5 km across Dubai Creek isn't 5 km of driving. Treat distance as a first filter, then check the commute in a maps app.

12 · Data goes stale fast LOW

Good listings rent within days. The file name carries the date; re-run before you call an agent.

13 · Be polite to the site MEDIUM

Keep the delay at 2 s or more and pages reasonable. Hammering the site increases the chance of a block for everyone on your connection.

14 · Bayut can change its site LOW

If both the JSON-LD block and the aria-labels change, parsing stops. The debug page + tests tell you quickly (chapter 15).

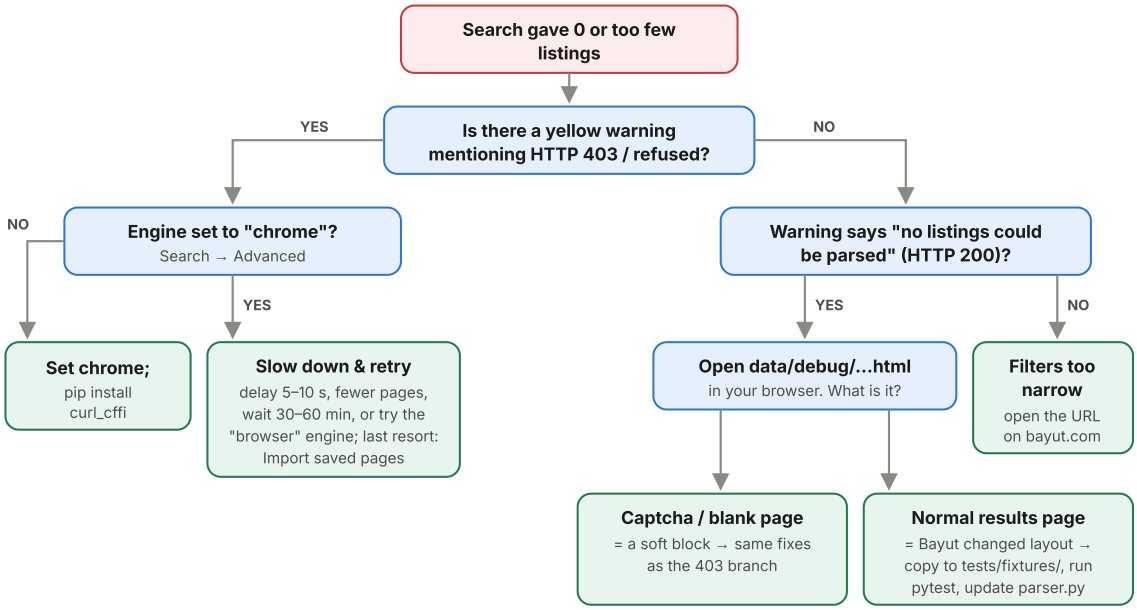
15 · Personal use, and keep raw pages private MEDIUM

Bayut's content belongs to Bayut and its advertisers, and listings include agents' contact details. Keep the data for your own search; don't publish or resell it. Raw saved pages also contain Bayut's internal tokens (GitHub blocked a push because of a Mapbox token in one). Don't commit raw pages; the test copy in `tests/fixtures/` was stripped of all scripts for that reason.

CHAPTER 15

Troubleshooting

Start at the top of the flowchart with what you see; follow the arrows to the fix.



FLOWCHART Diagnosing an empty or short search.

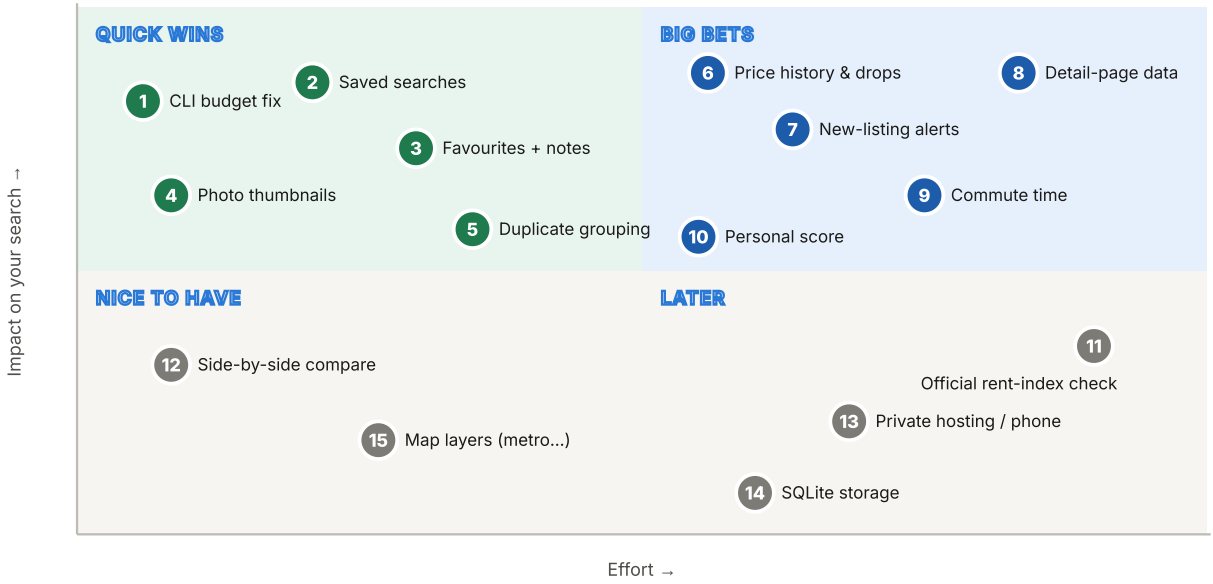
Symptom → cause → fix

You see	Likely cause	Fix
`No module named 'bayut_finder'`	Started from the wrong folder	`cd` into the folder that contains `Home.py`, then run again
`No module named 'curl_cffi'` / 'streamlit'	Different conda environment	`conda activate <env>` or `pip install curl_cffi`
`unexpected keyword argument 'width'`, `no attribute 'pills'`	Streamlit older than 1.50	`pip install -U "streamlit≥1.50"`
Yellow warning: refused (HTTP 403)	Bot check	Engine chrome; longer delay; wait; browser engine; import pages
Warning: HTTP 200 but no listings	Soft block or layout change	Open the file in `data/debug/` (flowchart)
Fewer listings than expected	Bayut ran out of results, filters narrow, duplicates removed	Open the search URL on bayut.com and compare the total count
Compare: "Needs both monthly and yearly..."	Only one rent period in the data	Search with both, or combine two datasets in the sidebar
Value score column empty	Fewer than 3 comparable listings	More pages, or a wider search
Map shows dots on a blank background	Map tiles blocked (VPN, firewall, offline)	Check internet access to openstreetmap.org; dots and table still work

CHAPTER 16

Next steps & enhancements

Fifteen ways to take the app further, placed by how much they'd help your search and how much work they are. Start in the top-left corner.



ROADMAP Impact vs effort. Numbers match the cards on the next pages.

Quick wins: an evening each

1 Monthly budget on the CLI too

Add `--budget-monthly` and convert per period exactly like the app, removing pitfall #2.

Touch: `bayut_finder/__main__.py` · reuse the conversion in `pages/1_Search.py`

2 Saved searches

Name a set of filters ("Marina 1–2BR ≤9k") and re-run it with one click. Stored as small JSON files next to `data/`.

Touch: `pages/1_Search.py` · `SearchParams.to_dict()` already exists

3 Favourites with notes & status

Star listings, add notes ("called, viewing Sat"), track status: new → contacted → viewed → offer. Survives new scrapes thanks to `listing_id`.

Touch: `st.data_editor` + a `favourites.csv`

4 Photo thumbnails in tables

The `image` column is already scraped. Show it as a small picture in each row to skim faster.

Touch: `bayut_finder/ui.py` → `st.column_config.ImageColumn`

5 Group duplicate listings

Same building + beds + sqft + price within 2% = probably the same flat. Show it once, with "posted by 3 agencies".

Touch: `analytics.py` (new `dedupe_key`), then medians and value score

Big bets: the most useful upgrades

6 Price history & price drops

Keep every scrape; for each `listing_id`, show first-seen date, days on market and price changes. A "↓ 5% this week" badge tells you who's ready to negotiate.

Touch: new `history.py` joining the CSVs (or SQLite, #14) · column in tables

7 New-listing alerts

Run your saved search every morning with Windows Task Scheduler (`python -m bayut_finder scrape …`) and send a Telegram or email message for new listings under budget.

Touch: small `alerts.py`, Telegram bot token, compare against the previous CSV

8 Detail-page enrichment

For shortlisted listings only, open the listing page and read furnishing, amenities, number of cheques, deposit, availability date and the description.

Touch: `parser.py` (new `parse_detail_page`) · keep it to ≤30 pages per run

9 Real commute time

Replace straight-line km with driving or transit minutes to your office, using OpenRouteService or OSRM (free tiers).

Touch: `pages/3_Map.py` · cache results per building

10 Your personal score

Sliders for what matters to you (price, size, distance or commute, verified, value score), combined into one ranked list. Example: 40% price, 25% commute, 20% size, 15% value score → "Top 10 for you".

Touch: new section in `pages/2_Compare.py` · weights saved with the search

Later and nice to have

11 Official rent-index check

Compare a listing's rent with official reference data where it's published, e.g. the rental index from Dubai Land Department / RERA, to see if an asking price is out of line.

Manual lookup link first; automate only if the source allows it

12 Side-by-side comparison

Pick 2–4 favourites and see them in columns: rent per month/year, size, AED/sqft, distance, value score, photo.

Touch: new page `pages/4_Shortlist.py`

13 Use it from your phone

Run the app on a home PC and open it from your phone over your Wi-Fi (`--server.address 0.0.0.0`). Cloud hosting is possible but private hosting is safer, and cloud IPs are more likely to be blocked.

No code change for local network use

14 SQLite instead of many CSVs

One small database file: faster loading, easy history (#6) and favourites (#3), no duplicate files.

Touch: `analytics.py` save/load functions only; pandas reads SQLite directly

15 Neighbourhood layers on the map

Toggle metro stations, schools, supermarkets, parks and beaches from OpenStreetMap, and add "nearest metro: 450 m" to each listing.

Touch: `pages/3_Map.py` · Overpass API, cached per area

⚠ One thing not to "improve": speed

Fetching many pages in parallel is easy to add, and the quickest way to get blocked. Keep requests sequential with a delay. If you need more data, spread runs over the day instead.

APPENDIX

File map, tests and glossary

Where to change what

File	Responsibility	Change it when…
`Home.py`	Landing page and tips	you want a different start page
`pages/1_Search.py`	Form, progress bar, saving, import tab	adding a filter to the form
`pages/2_Compare.py`	KPIs, charts, premium table, shortlist	adding a chart or ranking
`pages/3_Map.py`	Landmarks, geocoding, map, nearest table	adding landmarks or commute time
`bayut_finder/config.py`	Emirates, property types, periods, sort options, headers	Bayut adds a type or emirate slug
`bayut_finder/urls.py`	`SearchParams` + `build_search_url`	Bayut renames a URL parameter
`bayut_finder/fetcher.py`	chrome / browser / requests engines, retries, rotation	Bayut's bot check changes
`bayut_finder/parser.py`	JSON-LD + aria-label parsing and merging	Bayut changes its page layout
`bayut_finder/scraper.py`	Pagination, dedupe, debug pages, saved-page import	changing how many pages or stop rules
`bayut_finder/analytics.py`	Normalisation, value score, summaries, distance, storage	adding a metric or storage type
`bayut_finder/ui.py`	Shared filters, tables, colours	changing a filter on Compare *and* Map at once
`bayut_finder/__main__.py`	Command line	adding a CLI option

The safety net: 18 offline tests

```
> pip install pytest
> pytest
..................      [100%]
18 passed in 3.6s
```

They run against your saved Bayut page in `tests/fixtures/` and check: URL building for every filter; all 24 listings parsed with no missing fields; parsing still works with the JSON-LD removed, or with the aria-labels removed; monthly/yearly maths; value score; distances; pagination and stop rules; partial results when blocked; and the chrome engine against a local test server (403 → new fingerprint; empty page → debug file).

When Bayut changes: save a fresh results page over the fixture, run `pytest`, and the failing test names point to what moved.

Glossary

aria-label	A text label on a page element that describes it for screen readers ("Price", "Location"). Stable across Bayut deploys, so the parser relies on it.
JSON-LD	Hidden structured data (schema.org) that Bayut embeds for search engines; it holds clean numbers and GPS for every listing on the page.
TLS fingerprint	The pattern of a client's secure-connection handshake. Python and Chrome look different, which is how Bayut spots scripts.
curl_cffi	A Python HTTP library that can reproduce a real browser's TLS fingerprint (`impersonate="chrome"`).
Slug	The lowercase, hyphenated name in a URL: "Dubai Marina" → `dubai-marina` .
Median	The middle value when listings are sorted by price; less affected by outliers than the average.
Value score	"vs area median": a listing's % difference from the median of comparable listings. Negative = cheaper.
Monthly premium	How much more (or less) monthly listings cost over 12 months than yearly contracts in the same community and bedroom count.
Debug page	The raw HTML Bayut returned for a page that produced no listings, saved in `data/debug/` .